Deriving Strategic Insights from Earnings Calls: Leveraging Human Experts, AI-Based Search, and Generative AI for Synthesis and Summarization

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***Abstract*— Earnings transcript analyses have historically been processed manually and often augmented with AI techniques for natural language processing. Advances in Generative AI open up further opportunities for insights. Instead of looking in silos, we propose an integrated methodology that leverages human expertise, AI-based search, and Gen AI to improve the speed, accuracy, and relevance of insights along with enhanced explainability of Gen AI summaries. Our proposed methodology can augment executives with insights for informed decision-making in their business environment.**

Keywords—earnings calls, LLM, AI, financial disclosures

I. Introduction

Earnings calls are a critical medium through which publicly traded companies communicate their financial performance, strategies, and future outlook to analysts, investors, and other stakeholders. Companies also monitor earnings calls for their ecosystem - competitors, suppliers, and customers to better understand the impact of others' performance and common themes that are top of mind for analysts.

A typical earnings call features a Safe Harbor Statement (2-5 minutes), executive presentations (15-25 minutes), and an analyst Q&A session (30-40 minutes). The duration varies based on the company's size and market capitalization, with larger companies often having longer calls due to increased analyst interest. Most stocks are covered by 2-12 analysts, who usually ask 1-2 questions each during the Q&A. Earnings calls are held quarterly, and a 60-minute call typically contains 7,000-9,000 words. With around 4,300 publicly traded companies in the U.S., this results in an estimated total of 120-150 million words annually.

While these calls are rich with information, the presence of unstructured data (spoken and transcribed text), makes it difficult to efficiently extract meaningful insights. Traditionally, analysts have manually sifted through these transcripts, specialized software provides search capabilities, and there have been advances in text analyses in this area.

Recent advancements in artificial intelligence (AI) provide new opportunities for automating the analysis of large volumes of text. AI-based search algorithms can identify relevant passages quickly, while generative AI models can synthesize and summarize information, making it more accessible and actionable. This paper explores the combination of these technologies to derive strategic insights from earnings call transcripts, offering a potentially transformative approach to earnings transcript interpretation.

II. Literature Review

The financial industry has long acknowledged the value of earnings calls, leading to substantial research on methods for analyzing these transcripts. Traditional approaches often involve manual content analysis or simple keyword searches, which, while useful, are limited in handling the complexity and nuance of natural language.

Several studies have been conducted using a text analytics approach to studying corporate disclosures, earnings call transcripts and management discussion and analysis sections of corporate filings [1] . The text analysis methods employed range from manual analysis to computerized approaches. While manual approaches can yield precise findings related to specialized topic areas, they are limited in terms of costs, scalability, and replicability.

Most computerized approaches to analyzing financial text in the past two decades have used a dictionary-based approach that ranges from general dictionaries from social psychology literature to domain specific word lists based on financial documents such as earnings calls, that outperform the general wordlists in terms of predictive power [2],[3]. Subsequent research in this area has explored the use of machine learning models such as Bayesian machine learning model to extract tone information [4] and vector space models to evaluate changes in disclosure [5]. Other than dictionary-based approaches, most applications of NLP techniques in financial document analysis have focused on extracting topics and linguistic features from text such as sentiments, readability and evaluating its impact on price performance [6], [7].

In more recent research, we have seen the application of advances in deep learning technology to extracting insights from earnings calls, such as understanding digital strategy of companies [8] or as black box models for stock price performance predictions [9]. The ability of deep learning models to process a large variety of data has also enabled the use of such models for analysis of audio streams from earnings calls [10] as

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well as multi modal text and audio parings to derive deep insights[11].

The advent of LLM technology has seen interest in use of LLM's in the financial sector, where there are several applications for LLM for various financial information processing tasks [12]. Domain specific financial large language models (FinLLM's) have been developed by fine-tuning general purpose large language models with finance specific documents and datasets. The performance of FinLLM's has been tested and fine-tuned on a number of tasks such as sentiment analysis, classification, entity recognition, question answering, stock performance prediction and text summarization. However, researchers propose that FinLLM's can be extended and tested to perform more advanced tasks such as causality detection, numerical reasoning and structure detection in financial documents [12]. Causal analysis is another area where LLM's are particularly well suited for analysis and prediction [13].

While LLM's have substantial text analysis and synthesis capacity, to date most research has been limited to typical NLP tasks such as sentiment analysis [14]. More recent research has explored extracting higher-level information such as topics discussed and temporal orientation [15], however the generation capabilities of LLM's have not been extensively used. Despite advancements in machine learning and AI, there remains a gap in fully integrating AI-based search and generative AI to comprehensively process earnings calls. Most existing studies focus on one technology or the other, without fully exploring their combined potential.

III. Integrated Methodology

This section details the proposed architecture and process used to apply AI-based search and generative AI to earnings call transcripts. Large language models (LLMs) have advanced natural language processing, enabling automated analysis of extensive texts like earnings call transcripts. However, in our experiments, feeding entire transcripts into an LLM for summarization produced generic, context-lacking insights. This happens because LLMs struggle to maintain context over long texts, leading to vague summaries that miss strategic nuances. The paper suggests a more refined approach by combining human expertise with AI and GenAI. An overview of this architecture is shown in Fig. 1.

A. Data Preparation

The primary data includes publicly available earnings call transcripts from major companies across various industries and can potentially include other material public external facing communication and releases made. Transcripts are divided into sections, which are manually annotated by experts to highlight key topics, ensuring contextual understanding. This enables the foundational taxonomy and industry-specific synonyms. With advances in LLMs, future research could pave the way for automated paragraph segmentation, labeling, and synonym build.

B. Supervised Learning

Using the segmented and annotated data, supervised learning techniques are employed to train models on recognizing these topics within the text. This training process involves feeding labeled data into the model, allowing it to learn the patterns and context associated with different topics. The result is a model that can accurately identify and classify topics in earnings transcripts. The models can be leveraged for multiple quarter earnings transcript of a company and its ecosystem transcripts (competitors, suppliers, customers).

C. AI-Based Search

In response to a user query, AI-based search algorithms leveraging the learnings and semantic search identify relevant sections of the transcript more accurately rather than pure keyword matching. This extends across ecosystem transcripts providing easier compare and contrast for a query. The paragraph labeling provides the hyperlink to retrieve original documents for further exploratory understanding.

D. Generative AI Summarization and Conversation

For curated themes (analyst question categories, industry and sector trends, financials) and for ad-hoc conversations, the generative AI model is fed specific text chunks ensuring processing of manageable, contextually relevant segments, leading to more accurate and insightful synthesis and summaries. With the cost of GenAI becoming a priority for enterprises, future research can assess how such techniques can balance quality and relevance of outputs while optimizing speed and cost of Gen AI usage.

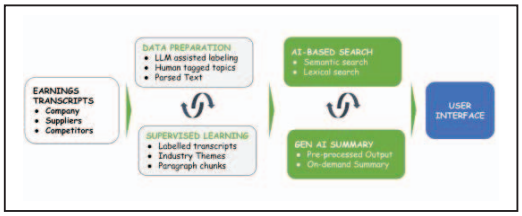

Fig. 1 AI Search and Gen AI Integrated Process Architecture

IV. Application Scenario

We demonstrate a real-world application of our proposed approach on earnings transcripts of Alphabet and Meta for fiscal Q3 and Q4 of 2023. Specifically, we leveraged the integrated methodology - human experts identifying primary and secondary topics relevant to financials, industry, and business model and utilizing AI powered Seach and Gen AI models that power search and summarization.

In Fig. 2, keyword search "investments" surfaces specific paragraph chunks with primary themes like "AI Infrastructure", "Cost Base Re-engineering" and additional company business model themes (Alphabet - "Mobile Content", Meta - "Reality Labs"). These insights are relevant to business executives, retrieved with speed, and hyperlinked text provides access to the original source.

The foundation of primary and secondary topics also is a key factor in Gen AI summaries - for accuracy and for explainability. Fig. 3 shows a prompt and response, along with the backend

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specific paragraph labels and text chunks used for the summary. While summaries enable executives to grasp the most important information quickly, the explainability adds to the confidence in any strategic decision-making.

Analyst Q&A is the longest section in earnings and given its dynamic nature, the most critical section for executives wherein analysts can ask any relevant question. Analysts often tend to cover common companies in an industry and have top themes as part of their coverage. Fig. 4 shows analyst firms and top themes and sub-themes for the software industry cohort.

Fig. 2. Results of keyword search "investments"

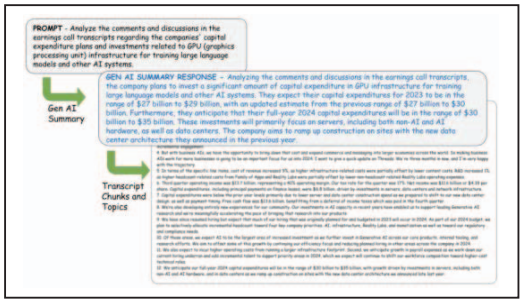

Fig 3. Gen AI Summary and Explainability

Goldman Sachs	**Ad** - *Brand Advertising, A* **AI** - *AI Theme, Open Sour* **Revenue** - *Revenue Growth* **Growth** - *H2 Growth, 202* **Brand** - *Brand Advertisin*
JP Morgan	**AI** - *Generative AI Tools,* **Ad** - *Returns On Investme* **Capex** - *Capex In '23, Cap* **Outlook** - *Outlook 1Q, Cap* **Revenue** - *Monthly Reven*
Morgan Stanley	**AI** - *AI Tools, Generative* **Ad** - *Advertising Business,* **Growth** - *AI Growth, Mix* **Revenue** - *Expenses Could* **2024** - *2024 Hiring, Rolls*
Bank of America	**Margin** - *TCn Margin, M* **Cost** - *Cost Of Learning,* **Growth** - *Cloud Growth, S* **Ad** - *Q4 Advertising Rev* **Q4** - *Q4 Guidance, Q4 Ex*
Barclays	**Ad** - *Ad Load, Ad Model,* **AI** - *Generative AI, AI Us* **Amazon** - *Amazon Deal, An* **Growth** - *Growth Rate, Us* **Meta** - *Meta's Foc, Meta*
Evercore	**Ad** - *AI Advertising, Long* **Growth** - *Growth In Maur,* **Google** - *Sinbs Google, Mo* **Amazon** - *Amazon Prime Vi* **AI** - *Youtube AI, AI Crea*
Lightshed Ventures	**Ad** - *Time Advertising Im,* **Video** - *Video Transition,* **Revenue** - *Revenue Decline* **Pinterest** - *Pinterest Ads* **AI** - *AI Targeting, AI Co*
UBS	**Monetization** - *Partner M, Ad* - *Advertising Media, S* **Partnership** - *Amazon Par,* **Amazon** - *Amazon Partner,* **Partner** - *Partner Monet*
Bernstein	**AI** - *My AI Usage, Llama I* **Meta** - *Meta Shopping, Mi,* **Ad** - *Advertisers Using, M* **Revenue** - *Revenue Guidan* **DR** - *Investment In DR, D*
Citi	**Growth** - *Brazil Growth, C* **Groups** - *Groups Reorgr,* **Business** - *New Business V,* **AI** - *AI Studio Timing, Ge* **Capex** - *Capex Spend, Ca*
MoffettNathanson	**Ad** - *Advertising Interest* **Factors** - *Big Wins Factor* **Adoption** - *Adoption Drive,* **Headwind** - *Headwind To ,* **Cost** - *Service Costs, Cos*
Wells Fargo	**AI** - *Generative AI Invest,* **Ad** - *Pinterest's Advertis* **Margin** - *Margin Guide, Ek* **Acceleration** - *1Q Acceler* **Fed** - *Fed Impact, Fed's*

Fig. 4. Analyst firms and top themes

In addition to themes, knowing what questions analysts are asking and how those are changing by fiscal quarter is an important area for executives. Gen AI summaries for Analyst Q&A and the ability to get deeper insights by themes can augment executives with faster and relevant insights. Fig. 5 shows results for query "AI and GenAI" for 2023 Q4 for Alphabet and Meta.

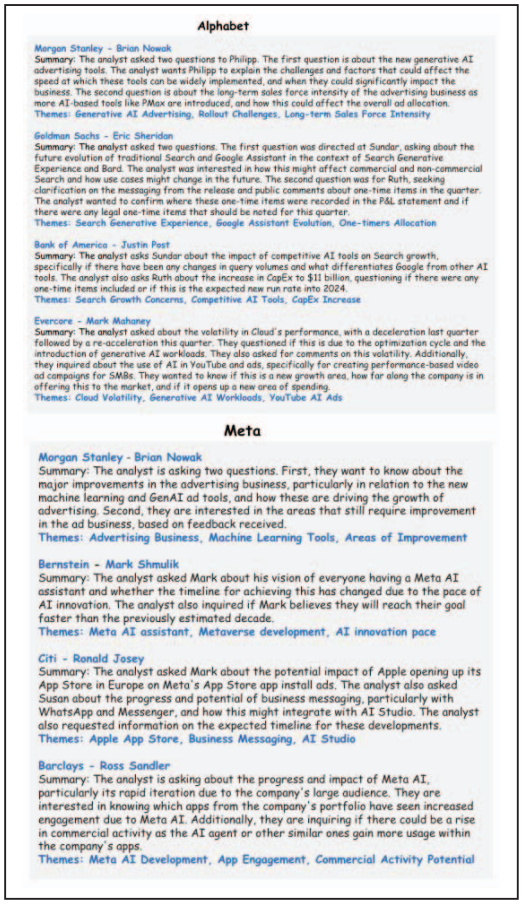

Fig. 5 Analyst QA Summary for "AI" and "GenAI" Topics

V. Conclusion

The use of Gen AI technology, while rapidly evolving, requires high quality data input and careful human prompting for effective use. Our proposed model that includes the use of human expertise, AI-based search, and generative AI for analyzing earnings call transcripts represents a practical approach for deriving substantial productivity benefits from Gen AI technology in the financial analysis domain, while accounting for the uncertainties inherent the outputs of evolving GenAI capabilities. This integrated methodology can improve the speed, accuracy, and relevance of insights and enhance confidence in Gen AI summaries through explainability of sources. As executives continue to navigate complex business environments, interpreting ecosystem earnings calls and knowledge of what matters to analysts will be crucial for maintaining a competitive positioning. Our proposed integrated methodology offers a potential approach to augment executives for informed decision-making in their business environment.

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Further research could focus on improving the interpretability of AI-generated summaries, as well as integrating additional data sources, such as social media sentiment or market trends, to provide an even richer analysis. As Gen AI technologies continue to evolve in areas of audio, their application in tonality and other speech dimensions of earnings call analysis is likely to expand, offering even more opportunities for insights and decision-making.

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